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Project Work Phase II – Review 1

DolFin : A Gamified Investment Learning and Portfolio Intelligence Platform for Women & Teenagers

By Batch 18

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INTRODUCTION



- In India, women's participation in direct investing remains very low, even though financial literacy has improved. Studies show this isn't because women don't understand investing - the real issue is a lack of confidence to actually start.
- Platforms like Groww, Zerodha, and ET Money are built for people who are already ready to invest. They help you execute trades, but they don't help you build the confidence to start in the first place. That's the gap.
- Women and teenagers have different needs when it comes to investing. Women often have specific goals like education or independence, and they learn better through hands-on experience than traditional lessons. Teenagers need a safe environment to practice before handling real money, but no platform is designed for learning by doing.
- DolFin uses AI-guided learning through gamified simulation. Users don't take lessons first - they start investing virtual money immediately. An AI agent watches every decision and teaches in real-time, explaining concepts like SIP, diversification, and risk exactly when users need them. This learning-by-doing approach builds confidence faster than traditional education.
- Users build complete practice portfolios, and the AI continuously analyzes their choices - flagging risks, suggesting improvements, and helping them understand what makes a good investment strategy before they ever use real money.



PROBLEM STATEMENT



PROBLEM STATEMENT:

Current investment education focuses on teaching concepts like mutual funds, SIPs, and diversification. Women complete these courses and understand the basics. But **70% still don't invest**. The gap between knowledge and action is where people get stuck.

Existing platforms like Groww and Zerodha are built for execution- they assume you're ready to invest. There is **no tool** that helps users build confidence through hands-on practice with real-time guidance before using real money.

Women and teenagers need a learning approach that's different from traditional lessons. Women have specific goals like education, travel, and independence. Teenagers need safe practice environments where they can make mistakes and learn without financial consequences. Generic platforms don't provide AI-guided learning-by-doing experiences.

Building an AI-coached practice platform can bridge the gap between understanding investing and having the confidence to actually do it.



Literature survey



Title of the Paper	Author Details and Published Year	Technique used	Shortcomings
Financial Literacy and its Influence on Youth Investment Decisions	Renu Bala Ningombam Jayanti(Sep 2025)	Cross-sectional survey with stratified random sampling (n=400) across regions. Analyzed using descriptive stats, correlation, regression, and SEM to study financial literacy's impact on youth investment behavior.	Cross-sectional design limits causal conclusions, self-reported data may introduce bias, sample may not fully represent India's diverse youth, and qualitative factors like family influence and cultural attitudes were not explored.
A Study of Demographic Factors and Investment Awareness Among Youth with Reference to Thane City	Dr. Sachin Vilas Acharekar, Dr. Santosh Tirathdas Karmani, CA Monica Lodha(2023)	Descriptive–explanatory study using convenience sampling (n=253) and a 5-point Likert questionnaire. Analyzed with frequency, percentage, Cronbach's Alpha (0.881), and Chi-Square (SPSS)..	Study is limited to Thane city only, small sample size, data based on perception only, and findings cannot be generalized to other cities.
Investment Behaviour Of Working Women	Tannu, Dr. Manoj Kumar Meet(Dec 2023)	This study is a literature review that analyzes existing research on working women's investment behaviour, focusing on factors such as returns, risk-taking, family influence, income, savings, and investment awareness.	No primary data; based only on secondary sources.No empirical/statistical testing, limited generalizability, and lacks post-COVID insights.



Literature survey



Title of the Paper	Author Details and Published Year	Technique used	Shortcomings
Assessment of Financial Literacy Among Working Indian Women	Akshita Arora(October 2015 – March 2016)	OECD-based questionnaire (16 items) administered to 444 working women in Rajasthan. Assessed financial knowledge, behaviour, and attitude using descriptive graphical analysis.	Study is limited to Rajasthan only, sample is biased towards educated urban women, and self-reported responses may not reflect true financial behaviour.
Financial Literacy and Investment Decisions among Youth: An Analysis Using Prism of Age	Dr. Munish Sharma, Anitika Damseth, Ishita Shyam(Oct 2024)	Mixed-method approach (online surveys, interviews, focus groups) with 631 respondents, using descriptive statistics, Chi-Square Test of Independence, and ANOVA/F-test via SPSS.	Study is limited to Himachal Pradesh only, uses self-reported data, and lacks longitudinal analysis to track changes in financial literacy over time.
FinTech Adoption and its Impact on Financial Literacy and Behaviour among College Students in Kerala	Dr. Janssen Joseph(Aug 2025)	Descriptive and analytical research using structured Likert-scale questionnaire (n=200), analyzed through descriptive statistics, Pearson correlation, and t-tests.	Study uses hypothesized data from only 200 students in Kerala, limiting generalizability, and does not establish a causal relationship between FinTech usage and financial literacy.



Literature survey



Title of the Paper	Author Details and Published Year	Technique used	Shortcomings
Financial Literacy of Finance Students: A Behavioral Gender Gap	Edina Berlinger, Barbara Dömötör, Krisztina Megyeri, György Walter (2025)	Survey-based analysis on 229 finance students using regression models and behavioral finance analysis	Study focused only on finance students in Hungary and mainly analyzed behavioral differences; lacked practical AI-based learning or simulation systems.
Improving Financial Literacy in College of Business Students: Modernizing Delivery Tools	Ronald Kuntze, Chen Wu, Barbara Ross Wooldridge, Yun-Oh Whang (2019)	Experimental study using online video-based learning modules and financial literacy tests on 244 business students	Study mainly focused on business students and online video learning; did not include AI-based personalization or real-time financial market analysis systems.
Formation of Financial Literacy in Primary School Students	Iryna Kuzma, Hanna Chaikovska, Iryna Levchyk, Oleksandra Yankovych (2022)	Experimental study using integrated financial education, media education, and sustainable development activities among 209 primary school students	The study was limited to a small sample from selected regions of Ukraine and mainly focused on school-level financial education without advanced digital financial tools or AI-based systems.



AIM & OBJECTIVES



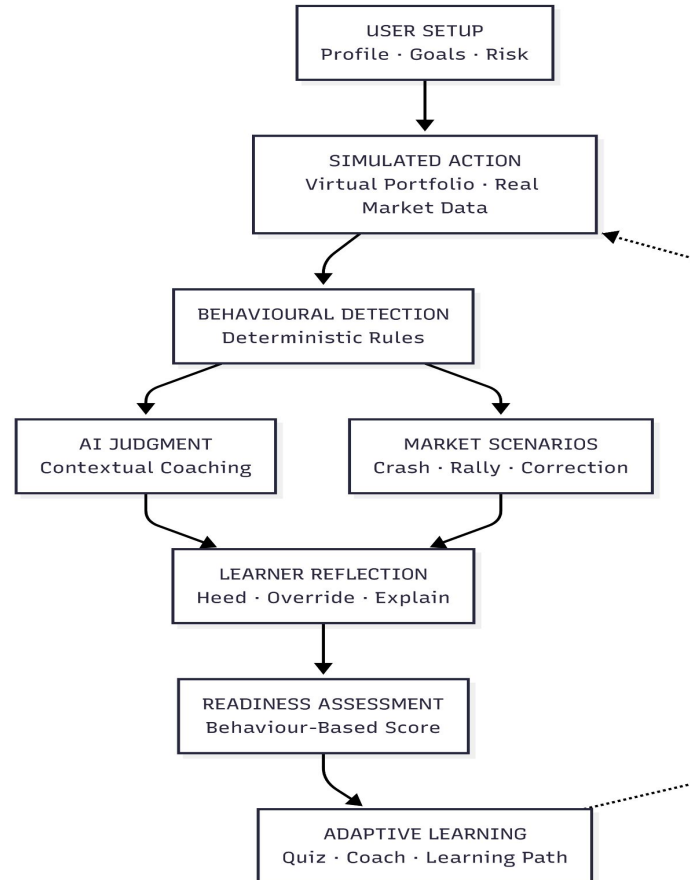
- **AI-Guided Learning:** Users learn by doing. AI explains concepts in real-time as users invest virtual money.
- **Gamified Simulation:** Users invest virtual money in mutual funds and stocks using Yahoo Finance data without financial risk.
- **Real-Time AI Coaching:** AI monitors every decision and provides instant feedback on risks, diversification, and fund choices.
- **Portfolio Analysis:** AI analyzes practice portfolios to identify mistakes, calculate risk scores, and suggest improvements.
- **Confidence Building:** Track user progress to help them know when they're ready to start real investing.



METHODOLOGY



Flow Diagram:





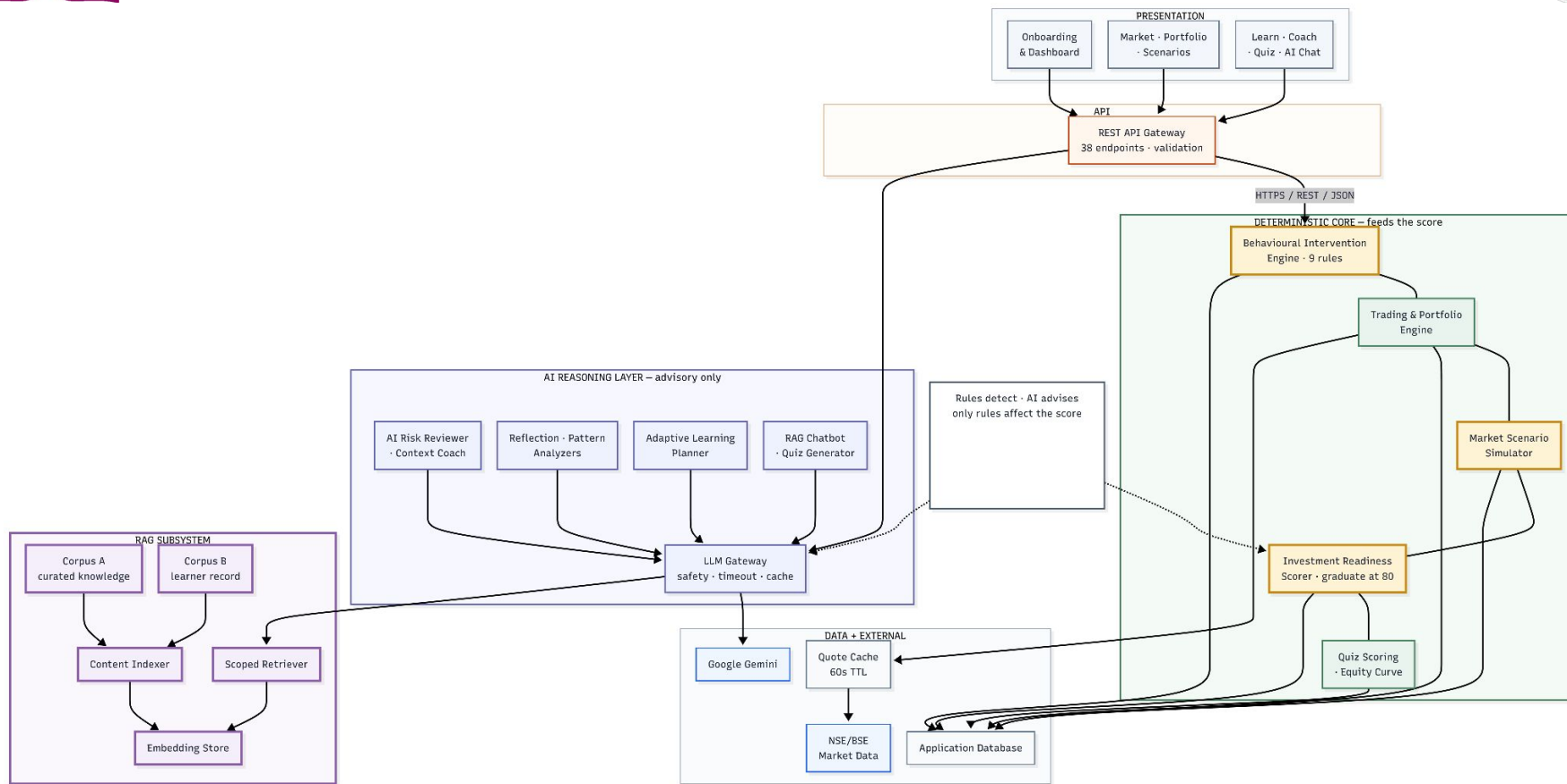
METHODOLOGY



- **User Setup:** Define profile, goals and risk preference
- **Market Simulation:** Practise investing with virtual money and real market data
- **Behavioural Detection:** Rules identify risky investment behaviour
- **AI & Scenarios:** Contextual guidance and simulated market conditions
- **Learning Reflection:** Learn from decisions, actions and explanations
- **Readiness Assessment:** Track behavioural progress and investment discipline
- **Adaptive Learning:** Personalised quizzes, coaching and learning paths

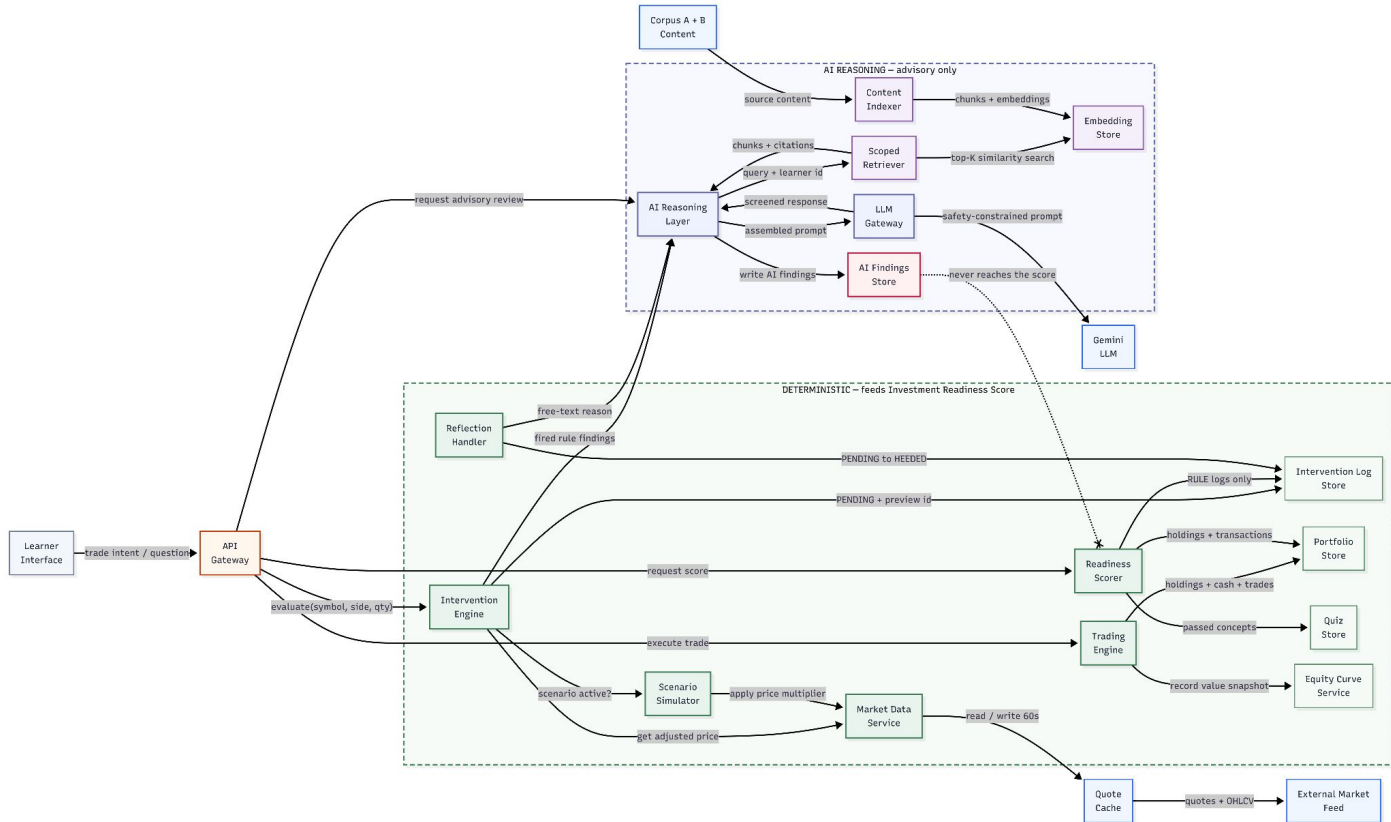


ARCHITECTURE OF THE SYSTEM



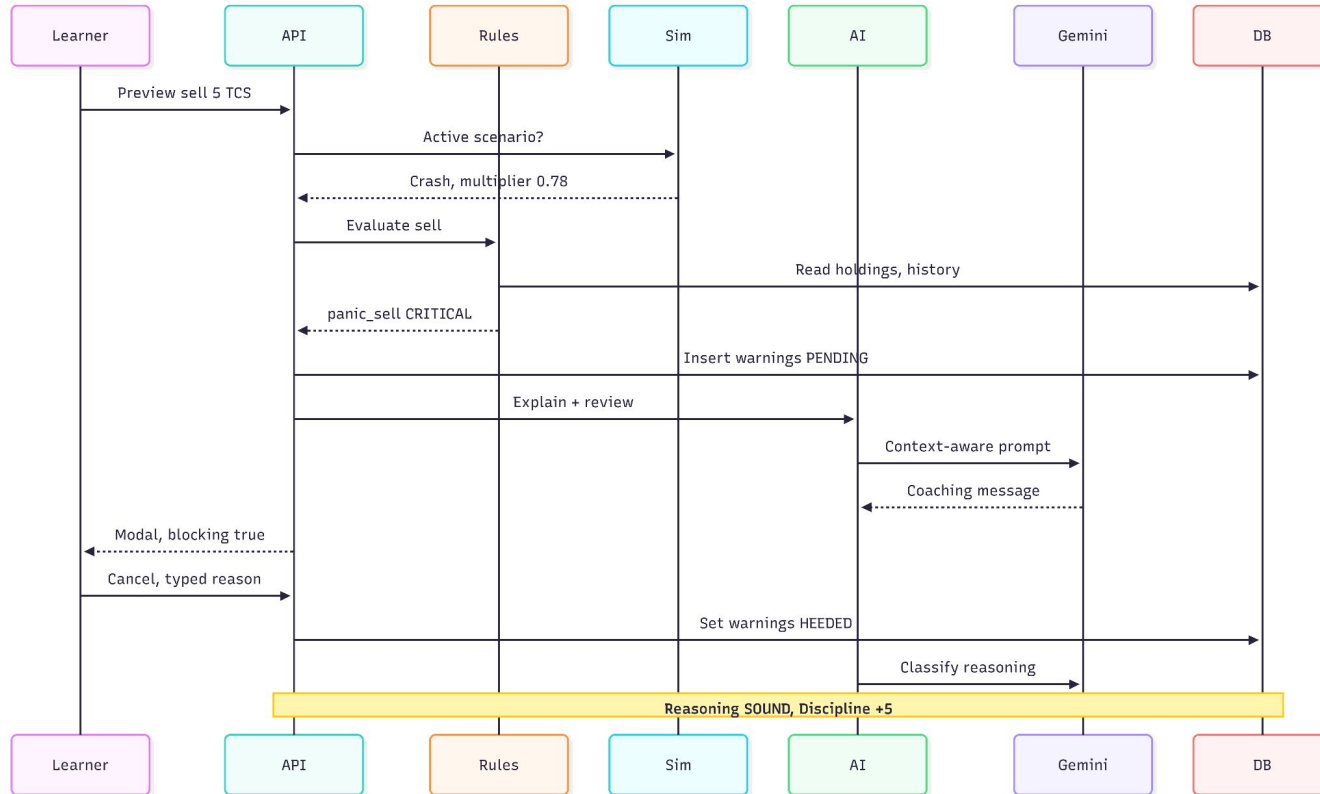


DESIGN DIAGRAMS



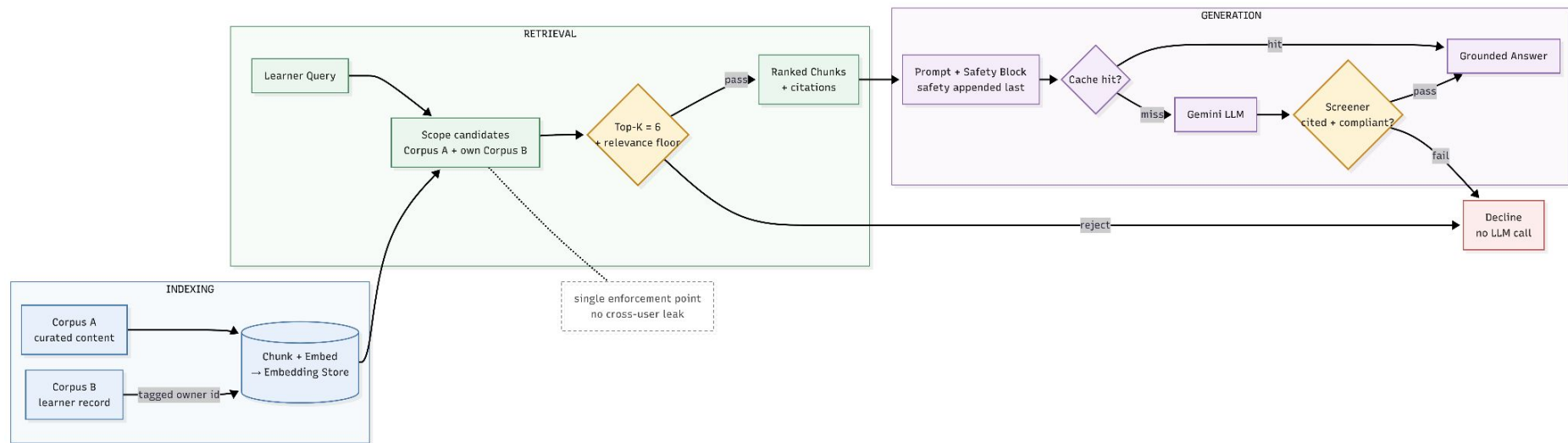


DESIGN DIAGRAMS





DESIGN DIAGRAMS





HARDWARE & SOFTWARE REQUIREMENTS



Hardware Requirements

- **CPU:** Intel i5 / AMD Ryzen 5 or better
- **RAM:** 8 GB minimum (16 GB recommended)
- **Storage:** 256 GB SSD minimum
- **Internet:** Required for real time stock data sync and updates

Software Requirements

OS: Windows 10/11, Ubuntu, or macOS

IDE: VS Code / PyCharm / Jupyter Notebook

Version Control: Git + GitHub

Language: Python 3.8+

Libraries:

- **Pandas** (data handling)
- **Flask / FastAPI** (backend)
- **Flutter** (mobile app)
- **Firebase** (database and authentication)
- **Scikit-learn** (AI portfolio analysis and risk scoring)
- **Plotly / Matplotlib** (visualization)
- **NSE/BSE datasets & yahooFinance API**



EXPECTED OUTCOME



Platform Delivers:

1. Safe Practice Environment

- Virtual ₹1L portfolio with real stock data (NSE/BSE)
- Experience market volatility without financial risk

2. AI Real-Time Coaching

- Instant feedback on investment decisions
- Behavioral interventions (panic prevention, risk warnings)
- Contextual learning (teach when user needs it)

3. Complete Skill Development

- Portfolio diversification & risk management
- Goal-based planning & progress tracking
- Emotional discipline & long-term thinking



PROJECT TIMELINE



Phase 1: Foundation & Setup (July)

- Development environment & Firebase setup
- Yahoo Finance API integration
- Backend structure (Flask/FastAPI)

Phase 2: Backend Development (August)

- User authentication & portfolio management
- Stock data fetching & caching
- Automated price updates (cron jobs)

Phase 3: AI/ML Engine (September)

- Rule-based intervention logic
- ML models (risk scoring, goal prediction)
- Behavioral pattern detection

Phase 4: Mobile Application (October)

- Flutter UI development
- Portfolio dashboard & stock browser
- Goal tracking & AI coaching interface

Phase 5: Integration & Testing (November)

- End-to-end integration
- Beta testing (30-50 users)
- Bug fixes & optimization

Phase 6: Finalization (December)

- Documentation & demo preparation
- Final review presentation



CONCLUSION



- Safe and interactive investment learning platform
- Improves financial literacy and confidence
- AI-guided and gamified learning experience
- Practical understanding of investments
- Bridges knowledge and investment action gap
- Real-world financial education potential



FUTURE SCOPE



- Add multilingual support to make the platform accessible to users from different regions of India.
- Add social and community learning features where users can participate in challenges, leaderboards, and peer discussions.
- Include voice-based AI assistance and chatbot support for users with limited financial or technical knowledge.
- Expand the platform for schools and colleges as a financial literacy learning solution for students.
- Community mentorship platform connecting beginners with finance experts.



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Queries?



Thank You